Shen Fu

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Research Interests

LLM serving and training, systems for MoE, GPU observability and performance analysis.

Education

University of Science and Technology of China Hefei, Anhui

Ph.D. in Computer Science and Technology *Sep 2024—Present*

- Transferred from M.S. program to Ph.D. program in 2026.
- GPA: 4.13/4.30 Advisor: Prof. Cheng Li
- Courses: Formal Languages and Complexity (100), Design and Analysis of Algorithms (98), Combinatorics (97), Advanced Computer Architecture (96)

University of Science and Technology of China Hefei, Anhui

B.E. in Computer Science and Technology *Sep 2020—Jun 2024*

- Admitted to USTC at age 16 through the *School of the Gifted Young*.
- GPA: 3.92/4.30 (Top 8%)
- Courses: Algebraic Structure (98), Basics of Artificial Intelligence (97), Compiler Principles and Techniques (96), Basics of Algorithms (94)

Research Projects

DeaMoE: Efficient MoE Architecture for Fast Small-Batch Decoding ADSL, USTC

Core Member *Dec 2025—Jan 2026*

- A novel MoE architecture using expert grouping and parameter sharing to significantly reduce expert weight loading per decoding step for small-batch workloads while preserving accuracy.
- Integrated the DeaMoE architecture into the training framework Megatron-LM and the inference framework vLLM.
- Developed custom inference operators for the two-stage routing strategy, achieving decoding speedup proportional to weight loading reduction on A40 and H100 GPUs.

SMIDT: Inference Framework for MoE with Dynamic Top-K Routing ADSL, USTC

Second Author, Core Member *Mar 2025—Aug 2025*

- An inference framework for MoE models with dynamic top-k routing. It automatically selects parallelism strategies to improve throughput on prefill-dominated workloads.
- Contributed to the model profiler, implemented support for dynamic top-k routing, enhanced pipeline parallelism, and helped design the parallelism planner.

Publications

[1] Zewen Jin, **Shen Fu**, Chengjie Tang, Youhui Bai, Shengnan Wang, Jiaan Zhu, Chizheng Fang, Ping Gong, and Cheng Li. 2026. SMIDT: High-Performance Inference Framework for MoE Models with Dynamic Top-K Routing. *Proceedings of the AAAI Conference on Artificial Intelligence* 40, 27 (March 2026), 22444–22453. https://doi.org/10.1609/aaai.v40i27.39403

Awards

- Qiangwei "Yuanzhi" Scholarship (**Top 3%**) — Oct 2023, USTC
- Jianghuai & NIO Automobile Scholarship — Jan 2023, USTC
- Chen Linyi Scholarship — Jan 2022, USTC
- Outstanding Freshman Scholarship (Silver) — Sep 2021, USTC

Miscellaneous

Professional Service

- Artifact Evaluation Committee Member, USENIX ATC '25

Teaching

- T.A. for *Compiler Principles and Techniques* (Instructor: Prof. Cheng Li) — Fall 2023, USTC

Open Source Contributions

- [sgl-project/sglang] feat: add dp attention support for Qwen 2/3 MoE models (#6121)

Skills

- **Languages**: Mandarin Chinese (Native), English (Fluent)
- **Programming**: Python, C/C++, Lua, Shell scripting
- **Frameworks**: PyTorch, vLLM, SGLang